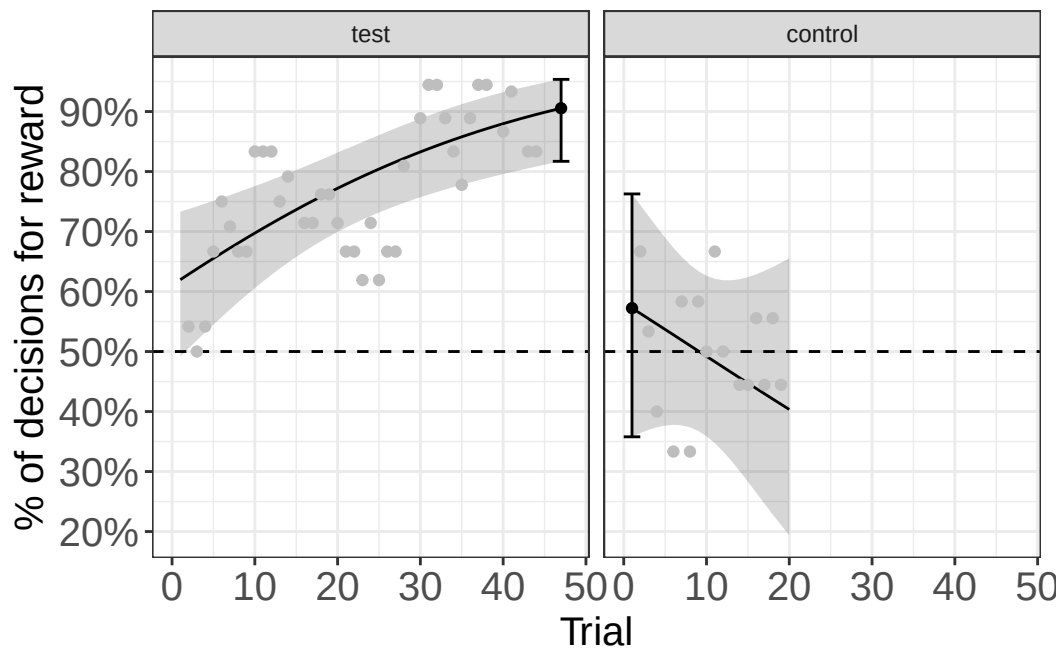


glm_analysis

Make the computations necessary for all questions:

Question 1: Can wasps use dorsal patterns?

- Looking at test versus control proportion correct
- Slope in test and control
- EMmeans at end of test and control
 - is this essentially equivalent to binning the last n trials and making the comparison



[1] "0.817-0.954"

	contrast	logOddsRatio	SE	df	asympt.LCL	asympt.UCL	z.ratio
1	test_end - control_start	1.968	0.555	Inf	0.881	3.055	3.547
	p.value	experiment	control_start	test_end	pFmt	confint	
1	0.000	perpPara_170725	1	47	<0.001	0.881-3.055	

During the test phase of the experiment, trial number (experience) had a significant positive effect on the expected probability (P_{corr}) of a correct decision (Figure 1 solid line) with every successive trial increasing the log odds of a correct decision by 0.043 (95% CI 0.019-0.067, $p < 0.001$).

The P_{corr} of a correct choice at the end of the test was higher ($P_{\text{corr}} = 0.905$, 95% CI 0.817-0.954) than at the start of the control ($P_{\text{corr}} = 0.572$, 95% CI: 0.358-0.763), and this difference was significant (log odds ratio = 1.968, $p < 0.001$) (Figure 1, compare error bars between test and control panel).

Also note that P_{corr} is significantly above chance (95% confidence ribbon does not overlap the 50% line) from an early trial number and remains that way until the end of the test. In contrast, the P_{corr} in the control phase is never significantly above chance for the duration of the control (95% confidence ribbon overlaps the 50% line).

That the model suggests naive wasps perform above chance at the onset of the test phase (ie. in trial 1) can be attributed to the fact that the model is unable to fit the rapid learning of the wasps, rather than to the availability and learning of olfactory and / or temporal cues during the pre-test training phase. This is supported by the proportion of correct decisions in the first few 3-trial wide windows of the test phase (black points) that are clustered around 50%.

Question 2: What information are wasps using to discriminate patterns?

- Do they need optic flow?
- Can they distinguish simple brightness differences?
- are there differences in performance among artificial stimuli?

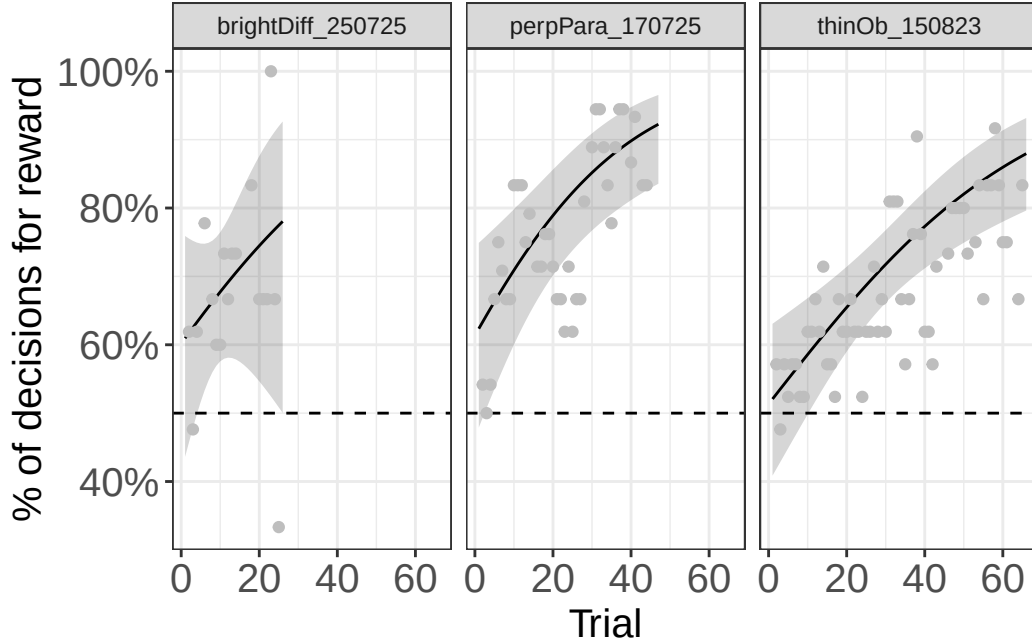


Table 1: Q2 trial-slope summary by artificial experiment

	logOdds	confint	pFmt	oddsMultiplier
brightDiff_250725	0.033	(-0.037, 0.103)	0.356	1.033
perpPara_170725	0.043	(0.019, 0.067)	<0.001	1.044
thinOb_150823	0.029	(0.015, 0.044)	<0.001	1.030

To answer whether wasps require optic flow information to discriminate patterns we tested whether they could distinguish between oblique stripes oriented at 45 degrees to the right or left of the y-maze arm’s long axis and the general direction of wasp travel.

Wasps had no trouble distinguishing between oblique stripes that presumably produce identical rates of optic flow in the dorsal visual field (Figure 2). Similar to the perpendicular-parallel experiment, trial had a significant positive effect on P_{corr} , with every successive trial increasing the log odds of a correct decision by 0.029 (95% CI (0.015, 0.044), $p < 0.001$; Table 1). Initially (at trial 1) performance was not significantly different from chance ($P_{\text{corr}} = 0.521$, 95% CI: (0.409, 0.631)) but rose steeply, reaching 0.880 (95% CI: (0.796, 0.932)) by the end of the experiment (trial 66), which is significantly higher than expected by chance (50%).

Next, we asked whether wasps could discriminate between an opaque and transparent roof, ie. between stimuli that carried little to no pattern information. While trial had a positive effect on P_{corr} (beta = 0.033, 95% CI (-0.037, 0.103)), unlike for the other two pairs of artificial stimuli this effect was not significant ($p = 0.356$), likely due to the much shorter number of

trials this experiment was run for. While statistical support for a positive slope is lacking, there is support for $P_{\text{corr}} > 50\%$ towards the middle and end of the experiment. At the onset of the experiment (trial 1), wasp performance was not significantly different from 50% ($P_{\text{corr}} = 0.610$, 95% CI: (0.436, 0.759)) but rises with experience and by the end of the experiment (trial 26) P_{corr} reached 0.780 (95% CI: (0.500, 0.927)). Performance at the end of the experiment is only borderline significantly higher than 50% because of the increased uncertainty in the slope / effect of trials on performance due to the lower number of trials run in this experiment. However, if one uses other metrics or analyses to compare performance to the 50% line, they are significantly different (for example, comparing performance at an intermediate trial or comparing the average performance of the last five trials to the 50% line).

NBNB! Slope or brightdiff exp not significant. fade best fit line?

Comparison among different artificial stimuli:

issues: - day is multicollinear in thin oblique experiment. It makes the slope of thin oblique appear less steep. This drives the difference in slopes detected between perppara and thin oblique. The issue is, is it real? I'm not confident enough that I ran the experiments in exactly the same way so as to make them comparable. If we remove day from the analysis, would we still find a significant difference in the slopes? Could the difference simply be attributable to slower learning due to the side bias which is clearly evident in the thin oblique experiment (possibly caused because of different pre-testing training regimes). Why even is there a side bias when these wasps had just come off the back of a previous experiment with a thick oblique stripe? was the side bias present at that time? - the issue im trying to get at above is that conclusions regarding slope diffs depend on the model you run. - massive confidence intervals around brightness experiment

Question 3: Can wasps use naturalistic patterns for navigation?

- Are performances different from 50%?
- Are performances different from the control?
- Are there differences in performance between naturalistic and artificial stimuli?

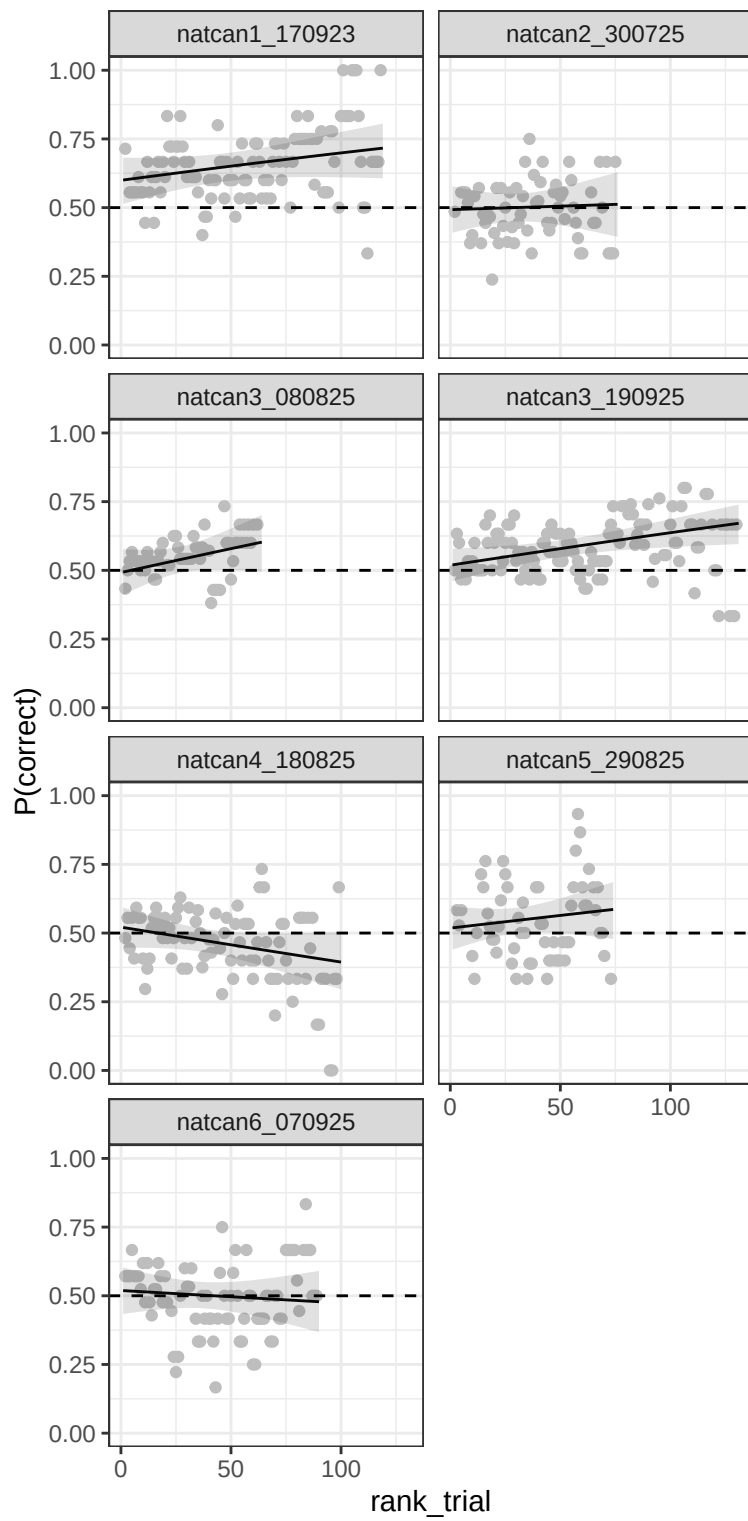


Table 2: Naturalistic test performance at first and last trial

experiment	startPcorr95CI	endPcorr95CI	endTrial	nat_or_art
natcan1_170923	0.600 (0.513, 0.682)	0.716 (0.606, 0.805)	119	naturalistic
natcan2_300725	0.493 (0.408, 0.577)	0.512 (0.394, 0.629)	76	naturalistic
natcan3_080825	0.494 (0.411, 0.577)	0.603 (0.495, 0.701)	64	naturalistic
natcan3_190925	0.520 (0.461, 0.578)	0.671 (0.595, 0.739)	131	naturalistic
natcan4_180825	0.520 (0.447, 0.593)	0.394 (0.296, 0.503)	100	naturalistic
natcan5_290825	0.519 (0.439, 0.598)	0.586 (0.479, 0.686)	74	naturalistic
natcan6_070925	0.519 (0.434, 0.603)	0.478 (0.368, 0.591)	90	naturalistic

Evidence for whether wasps can discriminate naturalistic canopies at above-chance rates ($>50\%$) is mixed. In four of the seven naturalistic canopy pairs we tested, the predicted P_{corr} by the end of the experiment was significantly above chance (the lower limit of the 95% CI is $> 50\%$, see Table x), suggesting wasps may be able to separate certain pairs of naturalistic canopies to some degree (assuming no other information was available to the wasps).

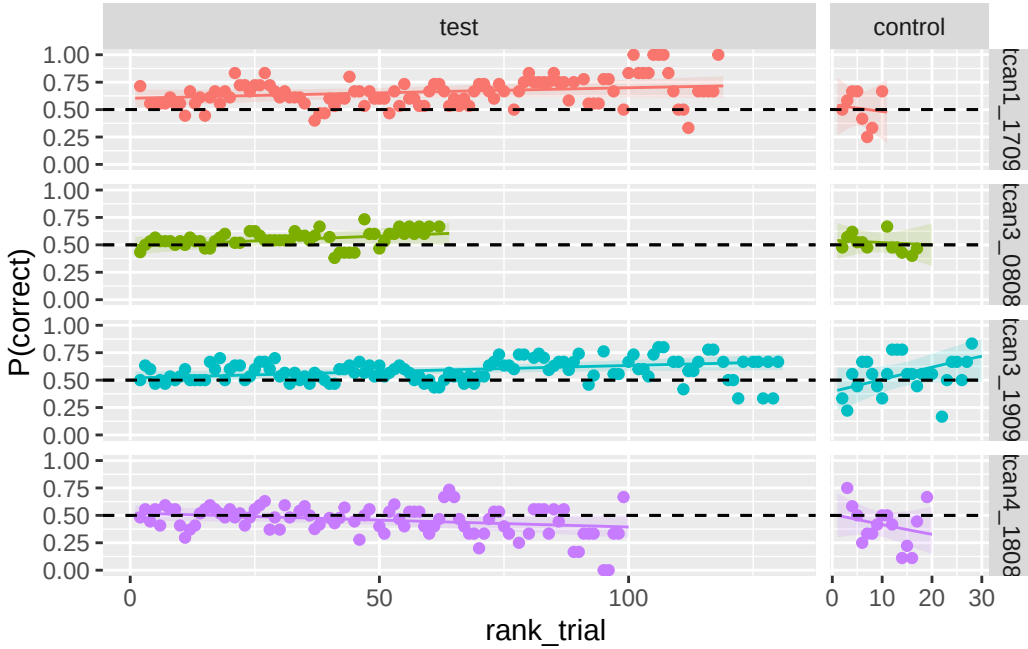


Table 3: Naturalistic test-control performance at test end and control start

experiment	testEndPcorr95CI	controlStartPcorr95CI	testEndTrial
natcan1_170923	0.716 (0.608-0.803)	0.538 (0.263-0.792)	119
natcan3_080825	0.604 (0.500-0.700)	0.540 (0.372-0.699)	64

experiment	testEndPcorr95CI	controlStartPcorr95CI	testEndTrial
natcan3_190925	0.667 (0.591-0.735)	0.406 (0.227-0.615)	131
natcan4_180825	0.394 (0.295-0.502)	0.500 (0.300-0.700)	100

naturalistic test vs control

nat vs art: Discrimination performance for naturalistic canopies tends to be lower than for artificial canopies (make some sort of comparison or just give ranges) and

Produce necessary tables to report from in-line and to include in MS.

Write results here: Evidence that wasps can discriminate naturalistic canopies from the test vs control analysis is also mixed.

Coming soon: 1. comparison of naturalistic stimuli versus artificial

ToDo: - make master tables - add end-trial estimates and CIs to all figures (q2 and q3 outstanding) - x and y figure axis labels - table headers and rownames need to be formatted for pub. - pub-ready experiment names? - eg. perpendicular / parallel and oblique (+45 - 45) - diffuser exp data in supplementary - naturalistic -> artificial transition experiment

Supplementary

Table: Test predictions at start, middle, and end trials

experiment	startPcorr95CI	midPcorr95CI	endPcorr95CI	mid, end trial
:-----	:-----	:-----	:-----	:-----
brightDiff_250725	0.61 (0.44, 0.76)	0.70 (0.58, 0.81)	0.78 (0.50, 0.93)	14, 26
natcan1_170923	0.60 (0.51, 0.68)	0.66 (0.60, 0.71)	0.72 (0.61, 0.81)	60, 119
natcan2_300725	0.49 (0.41, 0.58)	0.50 (0.45, 0.56)	0.51 (0.39, 0.63)	39, 76
natcan3_080825	0.49 (0.41, 0.58)	0.55 (0.50, 0.60)	0.60 (0.49, 0.70)	33, 64
natcan3_190925	0.52 (0.46, 0.58)	0.60 (0.56, 0.63)	0.67 (0.59, 0.74)	66, 131
natcan4_180825	0.52 (0.45, 0.59)	0.46 (0.41, 0.50)	0.39 (0.30, 0.50)	51, 100
natcan5_290825	0.52 (0.44, 0.60)	0.55 (0.50, 0.60)	0.59 (0.48, 0.69)	38, 74
natcan6_070925	0.52 (0.43, 0.60)	0.50 (0.45, 0.55)	0.48 (0.37, 0.59)	46, 90
perpPara_170725	0.62 (0.48, 0.75)	0.82 (0.73, 0.88)	0.92 (0.84, 0.96)	24, 47
thinOb_150823	0.52 (0.41, 0.63)	0.74 (0.68, 0.79)	0.88 (0.80, 0.93)	34, 66

Table: Test model coefficients

experiment	nat_or_art	term	logOdds	confint	pFmt
:-----	:-----	:-----	:-----	:-----	:-----
brightDiff_250725	artificial	(Intercept)	1.049	(0.101, 1.997)	0.030
brightDiff_250725	artificial	rank_trial	0.033	(-0.037, 0.103)	0.356
brightDiff_250725	artificial	sideAlt_trialTRUE	-0.381	(-1.212, 0.450)	0.369
brightDiff_250725	artificial	reward_sideR	-0.892	(-1.760, -0.024)	0.044
brightDiff_250725	artificial	sd_ (Intercept)	0.000	(NA, NA)	NA
perpPara_170725	artificial	(Intercept)	1.130	(0.385, 1.876)	0.003
perpPara_170725	artificial	rank_trial	0.043	(0.019, 0.067)	<0.001
perpPara_170725	artificial	sideAlt_trialTRUE	0.141	(-0.438, 0.721)	0.633
perpPara_170725	artificial	reward_sideR	-1.479	(-2.070, -0.888)	<0.001
perpPara_170725	artificial	sd_ (Intercept)	0.496	(NA, NA)	NA
thinOb_150823	artificial	(Intercept)	-1.122	(-1.709, -0.535)	<0.001
thinOb_150823	artificial	rank_trial	0.029	(0.015, 0.044)	<0.001
thinOb_150823	artificial	sideAlt_trialTRUE	0.021	(-0.464, 0.505)	0.933
thinOb_150823	artificial	reward_sideR	2.330	(1.801, 2.860)	<0.001
thinOb_150823	artificial	sd_ (Intercept)	0.000	(NA, NA)	NA
natcan1_170923	naturalistic	(Intercept)	-0.090	(-0.520, 0.341)	0.683
natcan1_170923	naturalistic	rank_trial	0.004	(-0.002, 0.010)	0.153
natcan1_170923	naturalistic	sideAlt_trialTRUE	0.204	(-0.161, 0.569)	0.273
natcan1_170923	naturalistic	reward_sideR	0.781	(0.416, 1.146)	<0.001
natcan1_170923	naturalistic	sd_ (Intercept)	0.159	(NA, NA)	NA
natcan2_300725	naturalistic	(Intercept)	-0.524	(-0.930, -0.118)	0.011
natcan2_300725	naturalistic	rank_trial	0.001	(-0.008, 0.010)	0.832
natcan2_300725	naturalistic	sideAlt_trialTRUE	0.048	(-0.315, 0.412)	0.794
natcan2_300725	naturalistic	reward_sideR	0.940	(0.581, 1.299)	<0.001
natcan2_300725	naturalistic	sd_ (Intercept)	0.122	(NA, NA)	NA
natcan3_080825	naturalistic	(Intercept)	0.388	(-0.031, 0.808)	0.070
natcan3_080825	naturalistic	rank_trial	0.007	(-0.004, 0.018)	0.195
natcan3_080825	naturalistic	sideAlt_trialTRUE	-0.211	(-0.573, 0.152)	0.255
natcan3_080825	naturalistic	reward_sideR	-0.629	(-0.992, -0.266)	<0.001
natcan3_080825	naturalistic	sd_ (Intercept)	0.089	(NA, NA)	NA
natcan3_190925	naturalistic	(Intercept)	-0.036	(-0.334, 0.262)	0.812
natcan3_190925	naturalistic	rank_trial	0.005	(0.001, 0.009)	0.013
natcan3_190925	naturalistic	sideAlt_trialTRUE	-0.034	(-0.287, 0.220)	0.795
natcan3_190925	naturalistic	reward_sideR	0.256	(0.010, 0.502)	0.042
natcan3_190925	naturalistic	sd_ (Intercept)	0.000	(NA, NA)	NA
natcan4_180825	naturalistic	(Intercept)	-0.114	(-0.474, 0.245)	0.533
natcan4_180825	naturalistic	rank_trial	-0.005	(-0.012, 0.001)	0.122
natcan4_180825	naturalistic	sideAlt_trialTRUE	-0.207	(-0.535, 0.121)	0.216

natcan4_180825	naturalistic	reward_sideR	0.608	(0.283, 0.933)	<0.001	
natcan4_180825	naturalistic	sd__(Intercept)	0.076	(NA, NA)	NA	
natcan5_290825	naturalistic	(Intercept)	0.597	(0.155, 1.038)	0.008	
natcan5_290825	naturalistic	rank_trial	0.004	(-0.005, 0.013)	0.412	
natcan5_290825	naturalistic	sideAlt_trialTRUE	-0.257	(-0.632, 0.118)	0.179	
natcan5_290825	naturalistic	reward_sideR	-0.792	(-1.168, -0.415)	<0.001	
natcan5_290825	naturalistic	sd__(Intercept)	0.000	(NA, NA)	NA	
natcan6_070925	naturalistic	(Intercept)	0.237	(-0.194, 0.667)	0.281	
natcan6_070925	naturalistic	rank_trial	-0.002	(-0.010, 0.006)	0.647	
natcan6_070925	naturalistic	sideAlt_trialTRUE	-0.168	(-0.559, 0.222)	0.398	
natcan6_070925	naturalistic	reward_sideR	-0.151	(-0.536, 0.234)	0.442	
natcan6_070925	naturalistic	sd__(Intercept)	0.000	(NA, NA)	NA	

Table: Test trial-slope summary

experiment	nat_or_art	logOdds	confint	pFmt	oddsMultiplier	
:-----	:-----	:-----	:-----	:-----	:-----	
brightDiff_250725	artificial	0.033	(-0.037, 0.103)	0.356	1.033	
perpPara_170725	artificial	0.043	(0.019, 0.067)	<0.001	1.044	
thinOb_150823	artificial	0.029	(0.015, 0.044)	<0.001	1.030	
natcan1_170923	naturalistic	0.004	(-0.002, 0.010)	0.153	1.004	
natcan2_300725	naturalistic	0.001	(-0.008, 0.010)	0.832	1.001	
natcan3_080825	naturalistic	0.007	(-0.004, 0.018)	0.195	1.007	
natcan3_190925	naturalistic	0.005	(0.001, 0.009)	0.013	1.005	
natcan4_180825	naturalistic	-0.005	(-0.012, 0.001)	0.122	0.995	
natcan5_290825	naturalistic	0.004	(-0.005, 0.013)	0.412	1.004	
natcan6_070925	naturalistic	-0.002	(-0.010, 0.006)	0.647	0.998	

Table: Test-control contrast summary

experiment	logOddsRatio	ci95	pFmt	controlStartTrial	testEndTrial	
:-----	:-----	:-----	:-----	:-----	:-----	
natcan1_170923	0.771	-0.494-2.036	0.232	1	119	
natcan3_080825	0.265	-0.538-1.068	0.518	1	64	
natcan3_190925	1.072	0.164-1.980	0.021	1	131	
natcan4_180825	-0.431	-1.381-0.519	0.374	1	100	
perpPara_170725	1.968	0.881-3.055	<0.001	1	47	

thinOb_150823	1.954	0.907-3.002	<0.001		1	66
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Table: Test-control predictions at test end and control start

experiment	controlStartPcorr95CI	testEndPcorr95CI	testEndTrial
:-----	:-----	:-----	-----:
natcan1_170923	0.538 (0.263-0.792)	0.716 (0.608-0.803)	119
natcan3_080825	0.540 (0.372-0.699)	0.604 (0.500-0.700)	64
natcan3_190925	0.406 (0.227-0.615)	0.667 (0.591-0.735)	131
natcan4_180825	0.500 (0.300-0.700)	0.394 (0.295-0.502)	100
perpPara_170725	0.572 (0.358-0.763)	0.905 (0.817-0.954)	47
thinOb_150823	0.492 (0.294-0.692)	0.872 (0.789-0.926)	66

Table: Test-control model coefficients

experiment	nat_or_art	term	logOdds	ci95	pFmt
:-----	:-----	:-----	:-----	:-----	:-----
brightDiff_250725	artificial	int	1.049	0.101-1.997	0.030
brightDiff_250725	artificial	trial	0.033	-0.037-0.103	0.356
brightDiff_250725	artificial	sideAltT	-0.381	-1.212-0.450	0.369
brightDiff_250725	artificial	reward_sideR	-0.892	-1.760--0.024	0.044
brightDiff_250725	artificial	sd__(Intercept)	0.000	NA-NA	NA
natcan1_170923	naturalistic	int	-0.090	-0.520-0.341	0.683
natcan1_170923	naturalistic	trial	0.004	-0.002-0.010	0.153
natcan1_170923	naturalistic	sideAltT	0.204	-0.161-0.569	0.273
natcan1_170923	naturalistic	reward_sideR	0.781	0.416-1.146	<0.001
natcan1_170923	naturalistic	sd__(Intercept)	0.159	NA-NA	NA
natcan2_300725	naturalistic	int	-0.524	-0.930--0.118	0.011
natcan2_300725	naturalistic	trial	0.001	-0.008-0.010	0.832
natcan2_300725	naturalistic	sideAltT	0.048	-0.315-0.412	0.794
natcan2_300725	naturalistic	reward_sideR	0.940	0.581-1.299	<0.001
natcan2_300725	naturalistic	sd__(Intercept)	0.122	NA-NA	NA
natcan3_080825	naturalistic	int	0.388	-0.031-0.808	0.070
natcan3_080825	naturalistic	trial	0.007	-0.004-0.018	0.195
natcan3_080825	naturalistic	sideAltT	-0.211	-0.573-0.152	0.255
natcan3_080825	naturalistic	reward_sideR	-0.629	-0.992--0.266	<0.001
natcan3_080825	naturalistic	sd__(Intercept)	0.089	NA-NA	NA

natcan3_190925	naturalistic	int	−0.036	−0.334−0.262	0.812	
natcan3_190925	naturalistic	trial	0.005	0.001−0.009	0.013	
natcan3_190925	naturalistic	sideAltT	−0.034	−0.287−0.220	0.795	
natcan3_190925	naturalistic	reward_sideR	0.256	0.010−0.502	0.042	
natcan3_190925	naturalistic	sd__(Intercept)	0.000	NA-NA	NA	
natcan4_180825	naturalistic	int	−0.114	−0.474−0.245	0.533	
natcan4_180825	naturalistic	trial	−0.005	−0.012−0.001	0.122	
natcan4_180825	naturalistic	sideAltT	−0.207	−0.535−0.121	0.216	
natcan4_180825	naturalistic	reward_sideR	0.608	0.283−0.933	<0.001	
natcan4_180825	naturalistic	sd__(Intercept)	0.076	NA-NA	NA	
natcan5_290825	naturalistic	int	0.597	0.155−1.038	0.008	
natcan5_290825	naturalistic	trial	0.004	−0.005−0.013	0.412	
natcan5_290825	naturalistic	sideAltT	−0.257	−0.632−0.118	0.179	
natcan5_290825	naturalistic	reward_sideR	−0.792	−1.168−−0.415	<0.001	
natcan5_290825	naturalistic	sd__(Intercept)	0.000	NA-NA	NA	
natcan6_070925	naturalistic	int	0.237	−0.194−0.667	0.281	
natcan6_070925	naturalistic	trial	−0.002	−0.010−0.006	0.647	
natcan6_070925	naturalistic	sideAltT	−0.168	−0.559−0.222	0.398	
natcan6_070925	naturalistic	reward_sideR	−0.151	−0.536−0.234	0.442	
natcan6_070925	naturalistic	sd__(Intercept)	0.000	NA-NA	NA	
perpPara_170725	artificial	int	1.130	0.385−1.876	0.003	
perpPara_170725	artificial	trial	0.043	0.019−0.067	<0.001	
perpPara_170725	artificial	sideAltT	0.141	−0.438−0.721	0.633	
perpPara_170725	artificial	reward_sideR	−1.479	−2.070−−0.888	<0.001	
perpPara_170725	artificial	sd__(Intercept)	0.496	NA-NA	NA	
thinOb_150823	artificial	int	−1.122	−1.709−−0.535	<0.001	
thinOb_150823	artificial	trial	0.029	0.015−0.044	<0.001	
thinOb_150823	artificial	sideAltT	0.021	−0.464−0.505	0.933	
thinOb_150823	artificial	reward_sideR	2.330	1.801−2.860	<0.001	
thinOb_150823	artificial	sd__(Intercept)	0.000	NA-NA	NA	

Table: Test-control trial-slope summary

experiment	nat_or_art	trialLogOdds	ci95	pFmt	oddsMultiplierPerTrial
:-----	:-----	:-----	:-----	:-----	:-----
brightDiff_250725	artificial	0.033	−0.037−0.103	0.356	1.033
perpPara_170725	artificial	0.043	0.019−0.067	<0.001	1.044
thinOb_150823	artificial	0.029	0.015−0.044	<0.001	1.030
natcan1_170923	naturalistic	0.004	−0.002−0.010	0.153	1.004
natcan2_300725	naturalistic	0.001	−0.008−0.010	0.832	1.001

natcan3_080825	naturalistic	0.007	-0.004-0.018	0.195	1.007
natcan3_190925	naturalistic	0.005	0.001-0.009	0.013	1.005
natcan4_180825	naturalistic	-0.005	-0.012-0.001	0.122	0.995
natcan5_290825	naturalistic	0.004	-0.005-0.013	0.412	1.004
natcan6_070925	naturalistic	-0.002	-0.010-0.006	0.647	0.998